The phosphosite distribution

How many sites each phosphoprotein contributed, and how that is spread over the dataset.

AT A GLANCE

Where	Phospho → Phosphosite Distribution
Data level	Phosphosite. The page reads the phospho matrix as it was uploaded; filtering and imputation performed elsewhere do not change it.
Needs	A phospho dataset carrying either a site_num column, or a protein column (Phosphoprotein, else Protein_group) together with PTM_Collapse_key.
One bar	One bar per site count: its height is the number of proteins carrying exactly that many sites.
Cutoff	An upper quantile trim, not a site count. See the pitfall.
Localisation	Ignored. Class I and Not Class I sites count alike.
Table	Yes, below the plot, downloadable as CSV.

AT PHOSPHOSITE LEVEL

The shared **Localization probability cutoff** selects sites across phosphosite views. At 0 it keeps all sites, including those with no probability; above 0 those unknowns are excluded. This is separate from completeness filtering and does not rerun normalization or imputation. An already collapsed phosphoprotein table has its own recorded collapse cutoff.

1 What the plot answers

The plot answers whether the phosphorylation in the dataset is spread thinly over many proteins or concentrated in a few. That is partly biology and partly method: a protein with forty sites is either a genuine signalling hub or a large, well-covered protein whose peptides the enrichment happened to like, and the shape of the distribution is what distinguishes a

dataset where such proteins are the tail from one where they are the body.

It is a frequency distribution, so both axes count proteins in different ways. Along the x axis is the number of sites a protein carries; up the y axis is how many proteins carry that number. The bar at $x = 1$ is therefore the number of proteins seen with exactly one site, not the number of sites seen once.

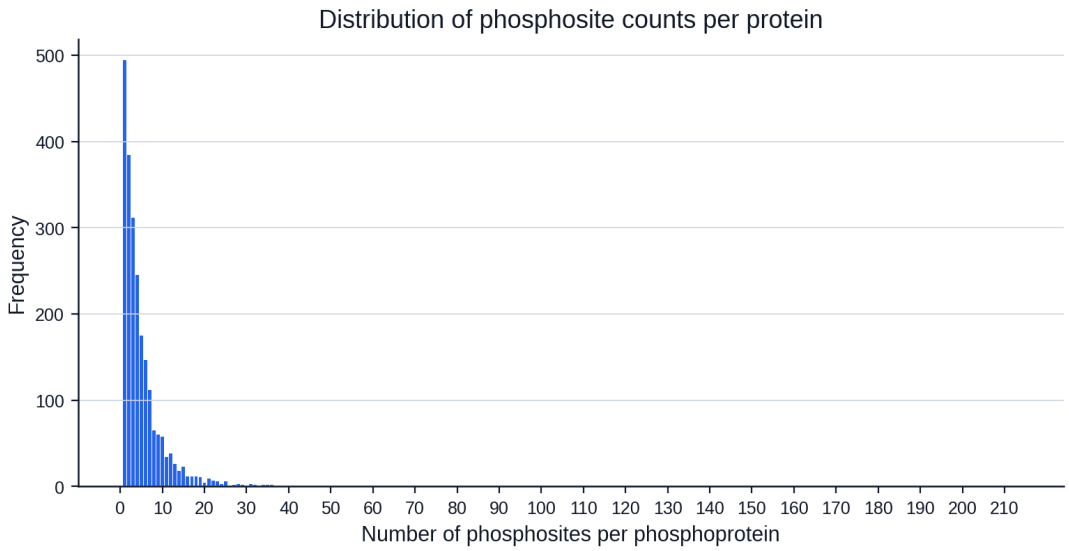


Figure 1.1. The distribution at **Cutoff** 0, the whole example dataset. The x axis is the number of phosphosites on a protein, the y axis how many proteins carry that many: 494 of the 2 301 phosphoproteins carry exactly one site, 384 carry two, and the frequency falls away steeply from there. The axis runs out to 213, the site count of the single most heavily modified protein — the tail to the right is a handful of proteins, not a large share of the data.

Reading it

- One bar: the number of phosphoproteins with exactly that many sites. The bars are drawn only at site counts that actually occur, so a gap on the axis means no protein in the dataset carries that number.
- The tail on the right: the heavily modified proteins. It is usually long and low, and the eye overstates it — read the y values before concluding that the dataset is dominated by them.
- The x ticks, placed every 10 sites once the largest count reaches 10, and every 1 below that. They are label positions and do not group the bars.

EASY TO GET WRONG

Cutoff is a quantile, not a number of sites. At 0.05 the tool takes the 95th percentile of the per-protein site counts and drops every protein above it — that is, the most heavily modified 5 % of proteins, whatever site count that turns out to be. It does *not* remove proteins with more than five sites. Ties at the quantile mean the removed fraction need not be exactly 5 %. This display trim is separate from the shared localisation-probability cutoff. 0 disables the trim altogether; the field is clamped to the range 0 to

1, and 1 would keep only the proteins at or below the minimum.

2 Controls

Controls on this page

Control	What it does	Default	Where
Cutoff	The fraction of the most heavily modified proteins to remove, as an upper quantile. 0 keeps everything; the field is clamped to [0, 1].	0	page
Width (cm), Height (cm), DPI	The size of the *downloaded* figure. The browser scales the rendered figure to fit the page.	20 × 15, 300	page
Download Plot	Exports the figure in the format chosen in the dropdown beside it: PNG, JPG, WEBP, SVG or PDF.	PNG	page

In the plot appearance panel

Control	What it does	Default	Where
Bar colour	The colour of every bar. Until one is chosen the plot draws its own sky blue, #87CEEB; the picker opens on #2563eb.	#87CEEB	appearance
Title, axis labels	Text, size, weight, colour and position, and whether the title is drawn. The plot writes Distribution of phosphosite counts per protein, Number of phosphosites per phosphoprotein and Frequency.	shown	appearance
Font	One face for the whole figure, or one per text.	plot default	appearance
Axis frame, grid, ticks	Which spines are drawn, the grid over them, and the size, rotation and colour of the tick labels.	default	appearance
Background, theme	Paper colour, transparency, and the preset themes.	default	appearance

WORTH KNOWING

The second table is the plot appearance panel, which is hidden until advanced mode is turned on (**Ctrl** + **Shift** + **E**). Everything it contains is described in Part XI; only the entries particular to this plot are repeated here.

3 How it is computed

HOW IT IS COMPUTED

The per-protein site count is obtained in one of two ways, and which one applies depends on the dataset rather than on any control:

- if the frame carries a site_num column — as a phosphoprotein-level frame does, one row per protein — that column is read as a number and rows where it is not one are dropped;
- otherwise the rows are grouped by Phosphoprotein, or by Protein_group where the first is absent, and the count is the number of *distinct* PTM_Collapse_key values in the group. This is the path an ordinary phosphosite upload takes.

With **Cutoff** above zero, the threshold is the $(1 - \text{cutoff})$ quantile of those counts and proteins whose count exceeds it are removed. The remaining counts are tabulated: for every site count k that occurs,

$$\text{frequency}(k) \;=\; \#\{\,\text{proteins with exactly } k \text{ sites}\,\},$$

and the pairs are drawn as bars, ordered by k.

Edge cases

- A protein whose sites were all removed upstream does not appear at $x = 0$; it is absent from the frame and so from the distribution entirely. There is no zero bar.
- The quantile is taken over whatever the previous step produced. On the site_num path that is one value per row, so if a frame ever carries site_num with one row per *site*, each protein enters the quantile once per site and the trim is weighted towards the heavily modified proteins it is meant to remove.
- Neither a site_num column nor a protein column with PTM_Collapse_key means no distribution can be built, and the page reports that rather than drawing an empty axis.
- If the trim leaves nothing, the figure is replaced by the words No phosphosite distribution data available.

- Localisation probability plays no part here, and neither do the sample values: a site counts towards its protein whether or not it was quantified anywhere.

4 The table below the plot

Columns in this table

Column	What it holds
SiteCount	A number of phosphosites per protein — one x position.
Frequency	How many phosphoproteins carry exactly that many sites — the height of that bar.

Only the site counts that occur have a row; the table has no zero rows to skip over, and it downloads as CSV.

5 In the notebook

A notebook starting point, using the frames introduced in *The function reference*. These compact examples do not encode every UI option; use the plot's notebook ZIP, where available, for its saved settings and preprocessing recipe:

PYTHON

```
fig = phossite_dist(data4, cutoff=0.01)
```

R

```
fig <- phossite_dist(data4, cutoff = 0.01)
```

cutoff is the same quantile trim as **Cutoff** on the page, here dropping the most heavily modified 1 % of proteins so that the tail does not stretch the axis. The function only takes the site_num path, so it wants the phosphoprotein frame data4 rather than the phosphosite frame; handed a frame without site_num it raises an error instead of grouping by protein as the page does.

6 Exporting

The download reproduces the plot at the width, height and resolution set on the page; the figure on screen is fitted to the window instead. PDF for a figure that will be scaled or edited, PNG for a slide. The table downloads separately as CSV.

Many datasets have a long right tail, but its shape is not a pass/fail criterion. More reported sites can reflect protein length, abundance, sequence coverage, enrichment or biology. A second mode does not identify its cause, and a short tail does not prove that enrichment failed. Keep the untrimmed table when discussing the proteins omitted from the display.

Behind this page

Function	What it produces
phossite_dist	The distribution, in the Python package
phospho_distribution_table	The two-column table under the plot, and the counting behind it
phospho_distribution_png	The figure the page draws